# FlutterShy User Manual and Theory

Alexander Ketzle

May 2026

# Contents

# 1 Introduction

This document comprises the user manual and theory explanation for *FlutterShy*. This documents expects the reader to be familiar with basic MATLAB programming in order to operate *FlutterShy* and expects the reader to become familiar with the underlying theory used for the calculations performed by *FlutterShy*.

# 2 User Manual

*FlutterShy* is intended to be used for high-power amateur rocketry by amateur rocketeers as a fin flutter prediction software. It is written in MATLAB and works on versions as old as R2023a. It requires user inputs on the physical properties of the fins, a RASAero II flight simulation file, and launch site altitude.

## 2.1 User Inputs

*FlutterShy* has a total of 14 parameters intended for users to manipulate; Nine correspond to physical properties of the fin, two correspond to the flight simulation data, and three are intended as advanced simulation controls that users should not need to adjust.

### 2.1.1 Fin Parameters

The first fin parameter is $c$, the average chord of the fin, in feet. This length is measured from the leading (forward) edge of the fin to the trailing (rearward) edge of the fin, parallel to the body of the rocket.

The second fin parameter is $m$, the mass per unit span of the fin, in slugs/foot. This parameter is calculated by taking the mass of the fin in slugs and dividing it by the span of the fin, in feet. Note the usage of the slugs unit; This is due to the atmospheric model script used in *FlutterShy*. One slug is a mass equivalent to 32.17405 lb in standard gravity.

The third fin parameter is $\overline{x}$ (in the code as x_bar), the chord-normalized distance of the center of gravity (c.g.) of the fin to the leading edge, a ratio value. This parameter is found by finding the location of the c.g. on the fin, then taking the chord-wise distance from the leading edge of the fin to the c.g. and dividing it by the chord at the span location of the c.g. on the fin. A value of 0 corresponds to a c.g. at the leading edge of the fin, a value of 0.5 corresponds to a c.g. centered between the leading and trailing edges, and a value of 1 corresponds to a c.g. located at the trailing edge. Most fins will have a value close to or exactly 0.5.

The fourth fin parameter is $\overline{r}$ (in the code as r_bar), the semichord-normalized radius of gyration of the fin, a ratio value. This value is found by taking the radius of gyration of the fin about the elastic axis and dividing it by the average semichord (half of the average chord).

The fifth fin parameter is $\omega_h$ (in the code as freq_h), the bending frequency of the fin in radians/second. This value can be found several ways. The author recommends either vibration testing the fin of interest under appropriate conditions, or performing modal analysis using the finite element analysis (FEA) suite of choice for the reader, assuming appropriate boundary conditions. A method of direct calculation can be found on page 40 of [1].

The sixth fin parameter is $\omega_\alpha$ (in the code as freq_alpha), the torsion frequency of the fin in radians/second. Just like $\omega_h$, this value can be found several ways. The author recommends either vibration testing the fin of interest under appropriate conditions, or performing modal analysis using the finite element analysis (FEA) suite of choice for the reader, assuming appropriate boundary conditions. A method of direct calculation can be found on page 40 of [1].

The seventh fin parameter is $a_h$, the chord-normalized distance of the elastic axis (e.a.) of the fin from the leading edge, a ratio value. This parameter is found similarly to $\overline{x}$, instead finding the location of the e.a., and taking the average value along the fin. A value of 0 corresponds to an e.a. at the leading edge of the fin, a value of 0.5 corresponds to an e.a. centered between the leading and trailing edges, and a value of 1 corresponds to an e.a. located at the trailing edge. Most fins will have a value close to or exactly 0.5.

The eighth and ninth fin parameters are $g_h$ and $g_\alpha$ (in the code as g_h and g_alpha respectively), the bending and torsional structural damping coefficients, respectively. These values are generally determined experimentally, however data does not exist for amateur rocketry applications. Values of 0.03 to 0.05 are

generally used in the aviation industry, however values in the range of 0.003 to 0.005 may be more appropriate for amateur rocketry.

### 2.1.2 Simulation Parameters

The first simulation parameter is the launch site sea level altitude in feet, which is in the code as site_altitude. This value is used to calibrate the atmospheric model used in calculations to provide accurate values derived from Mach number.

The second simulation parameter is the file path string to the RASAero II simulation data file in .csv format. This is in the code as RAS_Filepath. This is needed to provide the flight conditions needed for flutter calculation.

### 2.1.3 Advanced Simulation Controls

The controls presented here are intended to only be modified by advanced users who fully understand what these controls do and how they affect generated solutions. Most users should not have to change these controls from their default values. Changing these controls can significantly affect the performance and accuracy of *FlutterShy*.

The first advanced control is the size of the $1/k$ steps, in the code as invkstepsize. By default this value is 0.0001. This controls the fineness of the matrix generated for the $1/k$ method used for subsonic flutter solutions. Increasing the resolution (decreasing the magnitude of the value) from the default value causes a significant increase in *FlutterShy*'s runtime, while offering marginal accuracy increases. Reducing the resolution (increasing the magnitude of the value) will reduce *FlutterShy*'s runtime, at the cost of decreased accuracy, which could be significant in some cases.

The second advanced control is the maximum value of $1/k$ which *FlutterShy* tests to, in the code as invkMax. By default this value is 8. This controls the size of the matrix generated for the $1/k$ method used for subsonic flutter solutions. Increasing this value increases the size of the matrix generated by expanding the range of values for which the 1/k method is calculated. If *FlutterShy* is producing sensible results for subsonic flutter, modifying this value provides no benefit. The value should only be increased if flutter velocities seem extremely low for subsonic flight, as too small of a range can result in no intersection of the imaginary and real portions of the equations being generated, causing a value to be picked at an extrema that does not correspond physically. Generally, this value should not be increased past 14 unless necessary.

The third advanced control is the method handoff Mach number, in the code as machGate. By default this value is 1.01. This value controls the maximum flight Mach number at which *FlutterShy* will use subsonic methods instead of supersonic methods. By changing this value, the window of "subsonic" and "supersonic" flight is moved. The default value of 1.01 was selected due to the compressibility correction present in the Kearns supersonic method used by *FlutterShy*, which exhibits a singularity at Mach 1, or asymptotically approaches 0. It was found that the calculated values were generally stable starting near Mach 1.01, and thus at higher values corresponded to physical reality. Changing this value below 1 causes mathematical errors in calculation and produces imaginary flutter velocity numbers.

## 2.2 Outputs

When *FlutterShy* is run, it will generate a text output and several plots to inform the user.

### 2.2.1 Text Report

The text report generated by *FlutterShy* is intended to provide a "quick-view" or summary of the calculated results, highlighting information that users are likely to be most interested in. The report is broken into three sections:

1. Minimum Flutter Factor of Safety
2. Minimum Flutter Velocity
3. Maximum Flight Velocity

The first section provides the minimum factor of safety, the flutter velocity at the minimum factor of safety, the flight velocity at the minimum factor of safety, and the time in flight when the minimum factor of safety occurs. The second section provides similar information as the first, adjusted for the minimum flutter velocity. The third section provides the maximum flight velocity, the flutter velocity at that point, and the flutter factor of safety at that point.

### 2.2.2 Plots

*FlutterShy* generates eight plots for the user when calculation is complete. These plots are:

1. Flutter Factor of Safety vs. Time
2. Flutter Velocity vs. Sim Velocity
3. Flutter Mach vs. Sim Mach
4. Mach Numbers vs. Time
5. Flutter Factor of Safety vs. Dynamic Pressure
6. Flutter Factor of Safety vs. Altitude
7. Flutter Factor of Safety vs. Sim Mach
8. Flutter Velocity vs. Time

These plots are intended to give a more complete view of the flutter behavior of the vehicle in relation to several parameters. These plots present all of the calculated flutter values over the flight to apogee. Flutter velocity is not calculated after apogee because it is assumed that the vehicle deploys a recovery system at such point, and flutter will cease to be relevant. This saves computational time and resources.

# 3 Operating Theory

*FlutterShy* calculates flutter velocity based off of the $1/k$ method found in [1, 2] for subsonic flight and the Kearns supersonic method found in [3], which is a simplification of [4] using piston theory. *FlutterShy* does not perform any form of FEA, CFD, or FSI, nor does it use the SDPM or DLM methods. This allows it to provide flutter information quickly and with acceptable accuracy for amateur rocketry, however it will have reduced accuracy compared to the more complicated methods mentioned previously.

At a high level, *FlutterShy* ingests the user input fin parameters, launch site information, and RASAero II simulation data, generates a series of vectors for the flight conditions at each time step, calculates the flutter velocity using the appropriate method, and then bundles the data to provide the report and plots. In its current iteration, *FlutterShy* has only been tested to work with a mass ratio vector input to the two calculation methods, requiring scalars in other parameters. Further, the $1/k$ method function does not natively handle vectors of the mass ratio correctly, and thus requires *FlutterShy* to pass it mass ratio values as scalars in a for loop, significantly increasing computation time. Future updates hope to address these shortcomings to make *FlutterShy* both faster and more useful, potentially allowing for parameter spaces to be tested using the software.

Both the subsonic and supersonic method use the same set of fin input parameters for calculation, differing only on their use of either flight Mach number or reduced frequency. The common symbols are defined as:

1. $b$, Fin average semichord
2. $\mu$, the mass ratio
3. $\overline{x}$, the chord-normalized distance of the center of gravity from the leading edge
4. $a_h$, the chord-normalized distance of the elastic axis from the leading edge
5. $\overline{r}$, the semichord-normalized radius of gyration
6. $\omega_h$, $\omega_\alpha$, the respective bending and torsion natural frequencies of the fin
7. $g_h$, $g_\alpha$, the respective bending and torsion structural damping coefficients of the fin

furthermore, $\mu$ is defined in *FlutterShy* by

$$\mu = \frac{m}{\pi \rho b^2} \tag{1}$$

where $m$ is the mass per unit span of the fin, and $\rho$ is the air density of interest.

## 3.1 Subsonic Flutter Prediction Using the $1/k$ Method

The $1/k$ method was first described by Theodorsen and Garrick in NACA TR496 ([5]), then again in NACA TR685 ([1]), then cited by Fung in his book "An Introduction to the Theory of Aeroelasticity" ([2]). The revolves around $k$, the reduced frequency, which is defined by

$$k = \frac{\omega b}{V} \tag{2}$$

where $\omega$ is the frequency of oscillation, $b$ is the semichord of the fin, and $V$ is the airspeed. *Theodorsen's Circulation Function* (or simply *Theodorsen's Function*) $C(k)$ is defined in *FlutterShy* by

$$C(k) = \frac{H_1^{(2)}(k)}{H_1^{(2)}(k) + iH_0^{(2)}(k)} \tag{3}$$

where $H(k)$ are the Hankel functions of appropriate kind. Further, Theodorsen's function is broken into two parts by

$$C(k) = F(k) + iG(k) \tag{4}$$

equivalently

$$F(k) = \Re(C(k)) \tag{5}$$

$$G(k) = \Im(C(k)) \tag{6}$$

In *FlutterShy*, an array of $1/k$ and $k$ values is generated based upon the input range and resolution for $1/k$, then $C(k)$, $F(k)$, and $G(k)$ arrays are calculated based upon these arrays.

To find the flutter velocity, two quadratic equations are solved, one real and one imaginary, to find the point at which the roots of the two intercept when plotted against $1/k$. In *FlutterShy*, this is done by solving the quadratic equation for both equations at each point of interest. To do so, six components are calculated:

$$A_R = -(\mu + 1) - \frac{2G}{k} \tag{7}$$

$$A_I = \frac{2F}{k} \tag{8}$$

$$B_R = -(\mu\overline{x} - a_h) + \frac{2F}{k^2} - (0.5 - a_h)\frac{2G}{k} \tag{9}$$

$$B_I = \frac{1}{k}(1 + \frac{2G}{k} + (\frac{1}{2} - a_h)2F) \tag{10}$$

$$D_R = -(\mu\overline{x} - a_h) + (0.5 + a_h)\frac{2G}{k} \tag{11}$$

$$D_I = -(0.5 + a_h)\frac{2F}{k} \tag{12}$$

$$E_R = -(\mu\overline{r} + a_h^2 + 0.125) + (0.25 - a_h^2)\frac{2G}{k} - (0.5 + a_h)\frac{2F}{k^2} \tag{13}$$

$$E_I = \frac{1}{k}((0.5 - a_h) - (0.5 + a_h)\frac{2G}{k} - (0.25 - a_h^2)2F) \tag{14}$$

Then the real and imaginary equations are broken into three components each.
For the real equation:

$$\Delta_{R_A} = (1 - g_h g_\alpha)\mu^2\overline{r}^2\frac{\omega_h^2}{\omega_\alpha^2} \tag{15}$$

$$\Delta_{R_B} = \mu\frac{\omega_h^2}{\omega_\alpha^2}(E_R - g_h E_R) + \mu\overline{r}^2(A_R - g_\alpha A_I) \tag{16}$$

$$\Delta_{R_C} = A_R E_R - B_R D_R - A_I E_I + B_I D_I \tag{17}$$

Then for the imaginary equation:

$$\Delta_{I_A} = (g_h + g_\alpha)\mu^2\overline{r}^2\frac{\omega_h^2}{\omega_\alpha^2} \tag{18}$$

$$\Delta_{I_B} = \mu\frac{\omega_h^2}{\omega_\alpha^2}(g_h E_R + E_I) + \mu\overline{r}^2(A_I + g_\alpha A_R) \tag{19}$$

$$\Delta_{I_C} = A_I E_R - B_R D_I + A_R E_I - B_I D_R \tag{20}$$

The roots for the real equation are calculated by

$$X_{R_{1,2}} = \frac{\Delta_{R_B} \pm \sqrt{\Delta_{R_B}^2 - 4\Delta_{R_A}\Delta_{R_C}}}{2\Delta_{R_A}} \tag{21}$$

and the roots for the imaginary equation are calculated by

$$X_{I_{1,2}} = \frac{\Delta_{I_B} \pm \sqrt{\Delta_{I_B}^2 - 4\Delta_{I_A}\Delta_{I_C}}}{2\Delta_{I_A}} \tag{22}$$

unless $\Delta_{I_A} = 0$ (equivalently, $g_h = g_\alpha = 0$) then

$$X_{I_{1,2}} = -\frac{\Delta_{I_C}}{\Delta_{I_B}} \tag{23}$$

Complex roots of the real equation are ignored. Then the square root of every root is taken to produce the set of $\sqrt{X}$. This is useful because the following relation can be used to calculate flutter velocity later

$$\sqrt{X} = \frac{\omega_\alpha}{\omega} \tag{24}$$

equivalently,

$$\omega = \frac{\omega_\alpha}{\sqrt{X}} \tag{25}$$

Complex values of $\sqrt{X_I}$ are ignored. The values of $\sqrt{X}$ for each of the equations form horizontal parabolas when plotted against $1/k$. Due to the nature of the discrete calculation, the intercepts are located by finding the closest values between the real and imaginary roots, using

$$X_{ratio_1} = |1 - |\frac{\sqrt{X_{R_1}}}{\sqrt{X_{I_1}}}|| + |1 - |\frac{\sqrt{X_{R_1}}}{\sqrt{X_{I_2}}}|| \tag{26}$$

and

$$X_{ratio_2} = |1 - |\frac{\sqrt{X_{R_2}}}{\sqrt{X_{I_1}}}|| + |1 - |\frac{\sqrt{X_{R_2}}}{\sqrt{X_{I_2}}}|| \tag{27}$$

The minimum value for $X_{ratio_{1,2}}$ is found, and the two are compared, with the lowest determining the point used for flutter calculation. Using the above definitions of $\omega$ and $k$ at this point, *FlutterShy* calculates an initial flutter velocity using

$$V_{f_i} = \frac{\omega_\alpha b}{k\sqrt{X}} \tag{28}$$

*FlutterShy* then applies a compressibility correction to the flutter velocity. An initial Mach number, $M_{f_i}$ is calculated based on the speed of sound, $a$, at the point flutter is calculated, using

$$M_{f_i} = \frac{V_{f_i}}{a} \tag{29}$$

The corrected Mach number for subsonic flutter velocities is calculated using

$$M_{f_c} = \sqrt{M_{f_i}^2 \sqrt{1 - \frac{M_{f_i}^4}{4}} - \frac{M_{f_i}^2}{2}} \tag{30}$$

which comes from [1], and for supersonic flutter velocities the corrected Mach number is calculated using

$$M_{f_c} = \frac{1}{\sqrt{2}} \sqrt{M_{f_i}^2 \sqrt{4 + M_{f_i}^4} - M_{f_i}^2} \tag{31}$$

then the corrected flutter velocity is backed out using

$$V_{f_c} = M_{f_c} \times a \tag{32}$$

### 3.2 Supersonic Flutter Prediction Using the Kearns Supersonic Method

The Kearns supersonic method was presented in the paper "Flutter Simulation" ([3]) by J. P. Kearns, as a simplification of the supersonic method developed by Garrick and Rubinow in NACA TR846 ([4]). This method simplifies some of the supersonic aerodynamics using piston theory, producing a significantly simpler calculation method than outlined in the NACA report. However, this comes with some accuracy loss as well as the lack of any structural damping terms, creating a somewhat conservative estimate for flutter velocity.

Because of the use of piston theory, the method does not require further compressibility correction to be applied. In *FlutterShy*, the flight mach number, $M$ from the RASAero II simulation is pulled, then the equation is broken into three components for ease of calculation

$$A = \frac{\mu \overline{r}^2 \sqrt{M^2 - 1}}{\overline{x} b} \tag{33}$$

$$N = (1 - \frac{\omega_h^2}{\omega_\alpha^2})^2 + 4 \frac{\overline{x}^2}{\overline{r}^2} \frac{\omega_h^2}{\omega_\alpha^2} \tag{34}$$

$$D = 2(1 + \frac{\omega_h^2}{\omega_\alpha^2}) \tag{35}$$

which are then combined to calculate flutter velocity using

$$V_f = \omega_\alpha b \sqrt{A \frac{N}{D}} \tag{36}$$

# References


[1] Theodorsen, T., and Garrick, I. E., "Mechanism of Flutter A Theoretical and Experimental Investigation of the Flutter Problem," Tech. Rep. TR-685, NACA, 1940. URL https://ntrs.nasa.gov/citations/19930091762.

[2] Fung, Y. C., *An Introduction to the Theory of Aeroelasticity*, Dover, 1993.

[3] Kearns, J. P., "Flutter Simulation," Tech. rep., The John Hopkins University Applied Physics Laboratory, 1962. URL https://apps.dtic.mil/sti/citations/AD0650981.

[4] Garrick, I. E., and Rubinow, S. I., "Flutter and oscillating air-force calculations for an airfoil in two-dimensional supersonic flow," Tech. Rep. TR-846, NACA, 1946. URL https://ntrs.nasa.gov/citations/19930090942.

[5] Theodorsen, T., "General Theory of Aerodynamic Instability and the Mechanism of Flutter," Tech. Rep. TR-496, NACA, 1949. URL https://ntrs.nasa.gov/citations/19930090935.